# Conformal covariance of the supercritical Liouville quantum gravity metric

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**Abstract**

We prove conformal covariance for the canonical supercritical Liouville quantum gravity metric. The replacement for uniform convergence of metrics is a local gluing argument at endpoints surrounded by short loops. The proof uses lower-semicontinuous distance functions, compact confinement, local annular estimates for supercritical metrics, and a countable-scale endpoint regularity condition. No subcritical metric-convergence theorem is used.

## 1 Statement and conventions

Fix $\xi > \xi_{\mathrm{crit}}$ and write $Q = Q(\xi) \in (0, 2)$. Throughout, $D$ denotes one fixed normalization of the canonical metric assignment of Ding–Gwynne [1]. Metrics are allowed to take the value $+\infty$. If $W \subset \mathbb{C}$ is open, $D_h(x, y; W)$ denotes the intrinsic distance obtained by infimizing the $D_h$-length over paths contained in $W$; the infimum over an empty family of finite-length paths is $+\infty$.

**Theorem 1.1** (Conformal covariance)**.** *Let $U, V \subset \mathbb{C}$ be domains and let $\phi : U \to V$ be a deterministic conformal bijection. Let $h$ be a GFF plus a continuous function on $V$, in the local sense in which the canonical metric is defined. Then almost surely, simultaneously for all $u, v \in U$,*

$$D_h(\phi(u), \phi(v); V) = D_{h\circ\phi+Q\log|\phi'|}(u, v; U). \tag{1}$$

*The probability-one event is allowed to depend on $\phi$.*

Thus the assertion includes singular endpoints and the multiplicative constant is exactly one. We make no assertion here about one event holding simultaneously for all conformal maps.

For the proof it is more convenient to use the forward convention. Thus $h$ is a field on $U$ and

$$h^\phi = h \circ \phi^{-1} + Q \log |(\phi^{-1})'|, \tag{2}$$

and we write

$$d(u, v) = D_h(u, v; U), \qquad e(u, v) = D_{h^\phi}(\phi(u), \phi(v); V).$$

We first take $h$ to be the restriction of a whole-plane GFF to $U$.

Write $B_r(z) = \{w : |w - z| < r\}$ and $A_{a,b}(z) = B_b(z) \setminus \overline{B_a(z)}$. For a metric $D$, $D(\text{around } A)$ denotes the infimum of lengths of loops in $A$ separating its two boundary components, and $D(\text{across } A)$ denotes its crossing distance. Boundary endpoints are interpreted using paths in the closed annulus. The internal metric on a closed set is defined using paths contained in that set.

We use the following established inputs.

(I1) Locality, simultaneous continuous Weyl scaling, and covariance under complex affine maps for the canonical metric; see Ding–Gwynne [1, Definition 1.7, Theorem 1.8, Proposition 1.9].

(I2) Lower semicontinuity of the weak LQG distance function, local inverse-Hölder control, and existence of whole-plane geodesics between nonsingular points; see Pfeffer [2, Section 1.3, Propositions 1.10 and 1.12]. In particular, every finite metric-length path is Euclidean continuous.

(I3) Tight normalized annular distances, annular near-independence, and regularity estimates near geodesics from Ding–Gwynne [1, Sections 2–3].

(I4) The normalized GFF change-of-coordinates limit of Gwynne–Miller [3, Lemma 2.4]. This is a statement about fields and has no restriction involving $\xi$.

No metric-convergence theorem from the subcritical conformal-covariance proof is imported.

### 1.1 Compact path compactness

We record one consequence that will be used repeatedly.

**Lemma 1.2** (Compact path compactness)**.** *Let $K$ be a compact subset of a domain on which the field is locally a whole-plane GFF plus a continuous function. If $P_n : [0,1] \to K$ are paths with uniformly bounded metric lengths and are parametrized at constant metric speed, then $(P_n)$ has a Euclidean-uniformly convergent subsequence. If $P_n \to P$ uniformly, then*

$$\operatorname{len}(P; D_h) \le \liminf_{n\to\infty} \operatorname{len}(P_n; D_h).$$

*Consequently the metric intrinsic to a fixed Euclidean compact set is Euclidean lower semicontinuous.*

*Proof.* Cover $K$ by finitely many coordinate neighborhoods $W_j$ with $\overline{W_j}$ contained in slightly larger neighborhoods $W_j'$. On $W_j'$ write the field as a whole-plane GFF plus a continuous correction and extend the correction continuously, with a cutoff, to the plane. Locality identifies the lengths of subpaths contained in $W_j$ with the corresponding ambient lengths. The inverse-Hölder estimate for the ambient metric gives a common Euclidean modulus of continuity for constant-speed paths of bounded metric length; a finite cover yields equicontinuity on $[0,1]$. Arzelà–Ascoli gives a uniformly convergent subsequence.

For lower semicontinuity, subdivide $[0,1]$ so that each limiting subpath lies in one $W_j$. Uniform convergence puts the corresponding approximating subpaths in $W_j'$ for large $n$. Ambient lower semicontinuity on each piece gives the desired inequality after summing over the subdivision and then refining it. Applying the same argument to near-minimizing paths in a fixed compact set proves lower semicontinuity of its intrinsic metric. □

## 2 A preliminary comparison

**Lemma 2.1** (Bi-Lipschitz comparison)**.** *There exists a deterministic $C = C(\xi) \ge 1$, independent of $U$ and $\phi$, such that almost surely*

$$C^{-1} d(u,v) \le e(u,v) \le C d(u,v), \qquad u, v \in U. \tag{3}$$

*The same constant compares lengths of every path for the two metrics after the natural pullback.*

*Proof.* Set

$$m_{z,r} = r^{\xi Q} e^{\xi h_r(z)},$$

where $h_r(z)$ is the circle average. Fix $K \Subset U$. The source quantities

$$m_{z,r}^{-1} D_h(\text{around } A_{r,2r}(z)), \qquad m_{z,r} D_h(\text{across } A_{r/2,r}(z))^{-1}$$

are tight uniformly for $z \in K$ and $r$ small by affine covariance and the annular estimates of [1].

We next justify the same marginal tightness for the pulled-back metric without assuming the desired conformal covariance. Suppose otherwise. Then there are domains $U_n$, conformal maps $\phi_n$, points $z_n \in U_n$, and radii $r_n$ with $r_n / \operatorname{dist}(z_n, \partial U_n) \to 0$ along which the normalized estimate fails. Consider the normalized univalent maps

$$F_n(w) = \frac{\phi_n(z_n + r_n w) - \phi_n(z_n)}{r_n \phi_n'(z_n)}.$$

By the Koebe distortion theorem, $F_n \to \mathrm{id}$ and $F_n^{-1} \to \mathrm{id}$ locally uniformly, after passage to a subsequence, on every fixed disk on which they are defined. The convergence is in fact $C^\infty$ on compact subsets by Cauchy estimates. After subtracting a fixed smooth average from the rescaled field, the GFF change-of-coordinates input [3, Lemma 2.4] gives local total-variation convergence to the normalized GFF. The constant factor $\phi_n'(z_n)$ is removed exactly by affine covariance, including the $Q \log |\phi_n'(z_n)|$ term. The remaining continuous correction converges uniformly to zero on compact sets and hence changes all local lengths by factors tending uniformly to one by simultaneous Weyl scaling. Replacing the smooth normalization by the circle average costs a multiplicative log-normal factor with a law independent of $n$. This contradicts the assumed failure and proves uniform marginal tightness. Importantly, the compactness argument is over the normalized family of univalent maps, so the resulting bounds depend only on $\xi$.

Therefore, for each prescribed $p < 1$, there is $C_0 = C_0(p, \xi)$ such that, for all sufficiently small $r$ and all $z \in K$,

$$\mathbb{P}\big[e(\text{around } A_{r,2r}(z)) \le C_0 D_h(\text{across } A_{r/2,r}(z))\big] \ge p. \tag{4}$$

No independence of the two displayed distances is used. The same statement holds with $d$ and $e$ interchanged.

Choose $p$ close enough to one that the annular near-independence estimate, applied to geometrically separated radii in $[\varepsilon^2, \varepsilon]$, yields failure probability $O(\varepsilon^{10})$ at each lattice center. A lattice of mesh $\varepsilon^2/100$ has $O_K(\varepsilon^{-4})$ relevant centers. Hence, with probability tending to one, every point of $K$ is surrounded by a ball on which the loop-to-crossing comparison (4) holds. The events are local modulo an additive constant, so the annular near-independence estimate applies to the nonlinear pullback as well.

Let $P$ be a finite $d$-length path. Using successive exits from the good balls and concatenating the corresponding $e$-loops gives the standard planar loop-chaining construction of [2, Lemma 2.26]. The source crossing costs are charged to disjoint parameter intervals of $P$, so the total $e$-length is at most $C_0 \operatorname{len}(P; d) + o(1)$. The endpoints of the constructed path are only $O(\varepsilon)$ from those of $P$, but all paths lie in a fixed compact subset of $U$ and have uniformly bounded $e$-length. Lemma 1.2 therefore produces a limiting admissible path with the exact endpoints and no larger $e$-length.

Choose a deterministic sequence $\varepsilon_j \downarrow 0$ with summable failure probabilities and a countable compact exhaustion of $U$. On the resulting single event the argument applies to every finite-length path. Infimizing gives $e \le C_0 d$; interchanging the two metrics gives $d \le C_0 e$. Increase $C_0$ to at least one and call the result $C$. □

## 3 The tangent limit in the lower-semicontinuous topology

For nonnegative lower-semicontinuous functions, we use epigraph convergence: $f_n \to f$ if

(i) $f(x) \le \liminf_n f_n(x_n)$ whenever $x_n \to x$;

(ii) for each $x$ there exists $x_n \to x$ with $f_n(x_n) \to f(x)$.

For distance functions, $x$ is a pair of Euclidean points. We metrize local epigraph convergence using closed epigraphs on a countable exhaustion of $\mathbb{C}^2 \times [0,\infty]$. This gives a Polish realization of the relevant space of extended-valued lower-semicontinuous functions.

**Lemma 3.1** (Closed-set metric continuity under total variation)**.** *Fix closed disks $K \Subset K'$. Let $g_n, g$ be random generalized functions on a neighborhood of $K'$ which can be coupled so that $g_n = g$ on $K'$ with probability tending to one. Let $f_n, f$ be continuous functions on $K'$ with $\|f_n - f\|_\infty \to 0$ in probability. Then the intrinsic metrics on $K$ associated with $g_n + f_n$ converge jointly with the fields, in probability in the local epigraph topology, to the intrinsic metric associated with $g + f$.*

*Proof.* On the event $g_n = g$ on $K'$, locality identifies the metric functionals on $K$ before adding the continuous corrections. Simultaneous Weyl scaling gives, for every path $P \subset K$,

$$e^{-\xi\|f_n - f\|_\infty} \operatorname{len}(P; D_{g+f}) \le \operatorname{len}(P; D_{g+f_n}) \le e^{\xi\|f_n - f\|_\infty} \operatorname{len}(P; D_{g+f}).$$

After infimizing, the same comparison holds for the intrinsic distances on $K$. Since these intrinsic distance functions are lower semicontinuous by Lemma 1.2, uniform multiplicative convergence on finite values implies epigraph convergence, with $+\infty$ values preserved as well. □

**Lemma 3.2** (Expanding confinement)**.** *Let $h$ be a whole-plane GFF. For every fixed $R < \infty$ and every $\delta > 0$, there exists $S > R$ such that with probability at least $1 - \delta$, every $D_h$-distance between points of $\overline{B_R}$ is unchanged when paths are restricted to $\overline{B_S}$.*

*Proof.* Choose a large annular scale. By the short-loop barrier construction in Ding–Gwynne [1, Lemma 2.11 and its proof], with probability tending to one as the outer scale increases there exists a loop separating $B_R$ from infinity whose $D_h$-length is strictly less than half the cost of crossing a farther surrounding annulus. Any path between two points of $\overline{B_R}$ that exits that outer annulus hits the loop before and after the excursion; replacing the excursion by the shorter arc of the loop strictly decreases its length. Hence no distance-minimizing sequence needs to exit the outer disk. The same argument is valid when one or both distances are infinite. □

**Lemma 3.3** (Tangent limit)**.** *Suppose $U_n$ eventually contains every compact subset of $\mathbb{C}$, and $\psi_n : U_n \to V_n$ are conformal bijections with $\psi_n(0) = 0$ and $\psi_n'(0) \to a \neq 0$. Choose $R_n \uparrow \infty$ with $B_{2R_n} \subset U_n$, and set $K_n = \overline{B_{R_n}}$. Let $h$ be a whole-plane GFF and define*

$$d_n = D_h(\cdot,\cdot; K_n), \qquad e_n = \psi_n^* D_{h^{\psi_n}}(\cdot,\cdot; \psi_n(K_n)),$$

*extending each distance function by $+\infty$ outside $K_n \times K_n$. Then, jointly in the distributional topology for fields and the local epigraph topology for distance functions,*

$$(h, d_n, e_n) \Longrightarrow (h, D_h, D_h). \tag{5}$$

*Proof.* Fix a smooth radial test function $f$ of integral one and normalize $h$ by $(h, f) = 0$. Changing between fixed normalizations is harmless by Weyl scaling.

By the distortion theorem, $\psi_n \to a\,\mathrm{id}$ and $\psi_n^{-1} \to a^{-1}\,\mathrm{id}$ in $C^\infty$ on compact sets. Put

$$h^a = h(a^{-1}\cdot) - Q\log|a|.$$

The GFF coordinate-change lemma [3, Lemma 2.4], after subtracting the smooth average against $f_a(w) = |a|^{-2}f(w/a)$, gives local total-variation convergence of the normalized transformed fields to $h^a$. The restoration of the smooth average multiplies metric lengths by a random factor tending to one in probability. Moreover,

$$Q\log|(\psi_n^{-1})'| + Q\log|a| \longrightarrow 0$$

uniformly on compact sets, so the remaining Weyl correction has vanishing multiplicative effect.

Fix $R < S$. Restrict first to the metric intrinsic to $\overline{B_S}$. Under a maximal coupling of the normalized fields, they agree on a neighborhood of $\overline{B_S}$ with probability tending to one; Lemma 3.1 therefore gives joint convergence of the corresponding closed-disk metrics. Pullback by $\psi_n$ preserves epigraph convergence because $\psi_n$ and $\psi_n^{-1}$ converge uniformly on compact sets: the lower-bound condition is transported by forward convergence of endpoints, and recovery endpoints are transported by the inverse maps. Affine covariance identifies the limiting pulled-back metric with $D_h$.

It remains to pass from fixed confinement to $K_n$. Lemma 3.2 implies that for each fixed inner disk $\overline{B_R}$ and $\delta > 0$ one can choose $S$ so that, with probability at least $1 - \delta$, all distances between points of $\overline{B_R}$ agree with those confined to $\overline{B_S}$. Since $\psi_n(K_n)$ eventually contains every fixed target disk, the same barrier event transfers to the transformed field by the preceding total-variation comparison. Thus fixed-confinement convergence approximates expanding-confinement convergence uniformly in probability on each inner disk. This proves tightness and identifies every subsequential marginal limit of $d_n$ and $e_n$ as $D_h$.

Finally, in any subsequential joint limit the transformed field is the deterministic affine transform $h^a$ of the original field. The metric is a measurable functional of that field. Since the affine transform is invertible and affine covariance identifies its pullback with $D_h$, both metric coordinates in the joint limit are the same metric associated with the same $h$. Hence every subsequential joint limit equals the law in (5). □

## 4 Recovery at regular endpoints

Let

$$\Delta = \{2^{-j} : j \in \mathbb{N}\}.$$

The use of a countable scale set is deliberate: it is sufficient for the endpoint gluing argument and makes the later failure events universally measurable without an uncountable quantifier over radii.

**Lemma 4.1** (Deterministic recovery lemma)**.** *Let $d_n, e_n$ be Euclidean-lower-semicontinuous length metrics on sets eventually containing every compact subset of $\mathbb{C}$, extended by $+\infty$ outside those sets. Assume that finite-length paths are Euclidean continuous,*

$$C^{-1}d_n \le e_n \le Cd_n,$$

*and both sequences converge locally in the epigraph topology to the same metric $d$.*

*Suppose $u_n, v_n$ lie in a fixed compact set, $|u_n - v_n| \ge b > 0$, and*

$$L_n = d_n(u_n, v_n) \le M < \infty.$$

*Assume there are $\theta > 0$ and a sequence $s_j \downarrow 0$ such that, for $x = u_n, v_n$, every fixed $j$, and all sufficiently large $n$,*

$$d_n(x, \partial B_{s_j}(x)) \le M s_j^\theta, \qquad d_n(\text{around } A_{s_j,\sqrt{s_j}}(x)) \le M s_j^\theta. \tag{6}$$

*Then there is no subsequence on which*

$$e_n(u_n, v_n) \ge (1+\eta) L_n$$

*for a fixed $\eta > 0$.*

*Proof.* Pass to a subsequence with $u_n \to u$, $v_n \to v$, and $L_n \to L$. Epigraph lower semicontinuity gives

$$0 < d(u, v) \le L \le M,$$

where positivity follows from $|u - v| \ge b$ and the fact that $d$ is a metric on nonsingular separated endpoints whenever the displayed finite upper bound holds.

Choose recovery endpoints $a_n \to u$, $b_n \to v$ for $e_n$ and paths $Q_n$ from $a_n$ to $b_n$ whose $e_n$-lengths converge to $d(u, v)$. Fix $j$ so large that the two annuli around $u_n$ and $v_n$ are disjoint. For large $n$, $a_n \in B_{s_j}(u_n)$ and $b_n \in B_{s_j}(v_n)$. Around $u_n$, choose a loop in $A_{s_j,\sqrt{s_j}}(u_n)$ of $d_n$-length at most $2Ms_j^\theta$ and a path from $u_n$ to $\partial B_{\sqrt{s_j}}(u_n)$ whose initial segment to $\partial B_{s_j}(u_n)$ has $d_n$-length at most $2Ms_j^\theta$. The path crosses the loop. Hence comparison gives an $e_n$-connector from $u_n$ to every point of the loop of cost $O(CMs_j^\theta)$. Do the same at $v_n$.

By Euclidean continuity, $Q_n$ crosses both loops. Splice the middle part of $Q_n$ with the two connectors. Then

$$\limsup_n e_n(u_n, v_n) \le d(u, v) + O(CMs_j^\theta).$$

Let $j \to \infty$. If $e_n(u_n, v_n) \ge (1+\eta)L_n$ along the chosen subsequence, then

$$(1+\eta)L \le d(u, v) \le L,$$

a contradiction. □

Let $\theta > 0$ be the exponent supplied by the regular-geodesic estimates of Ding–Gwynne [1, Section 3.2]. Fix $\alpha \in (3/4, 1)$. We call $u \in \partial B_{\alpha r}(z)$ and $v \in \partial B_r(z)$ a *dyadically regular pair* if they are joined by a $D_h$-geodesic in $A_{\alpha r, r}(z)$,

$$D_h(u, v) \le (1-\alpha)^{-1} m_{z,r}, \tag{7}$$

and, for $x = u, v$ and each $\delta \in \Delta$ with $0 < \delta \le (1-\alpha)^2$,

$$\max\left\{ D_h(x, \partial B_{\delta r}(x)), D_h(\text{around } A_{\delta r, \sqrt{\delta} r}(x)) \right\} \le \delta^\theta D_h(u, v). \tag{8}$$

### 4.1 Measurability of failing regular pairs

**Lemma 4.2** (Universal measurability)**.** *Fix a compact annulus $A$, constants $M, b > 0$, and $\eta > 0$. On the Polish product space consisting of two lower-semicontinuous length-distance functions on $A$, together with endpoints and a continuous path, the set of tuples for which the path witnesses a dyadically regular pair of source distance at most $M$, endpoint separation at least $b$, and comparison failure*

$$e(u, v) \ge (1+\eta) d(u, v)$$

*is Borel after the path is included as a coordinate. Its projection to the two metric coordinates is analytic and hence universally measurable. On that projected event there exists a universally measurable choice of witnessing endpoints; a witnessing path may also be selected, but it will not be needed in the probabilistic subsequence argument.*

*Proof.* Use $C([0,1],A)$ with the uniform topology. Endpoint conditions, Euclidean containment conditions, and membership in the countable family of dyadic annuli are Borel. For lower-semicontinuous length metrics, inequalities involving distances between specified endpoints are Borel under epigraph convergence. Path-length sublevel conditions are Borel after expressing length as the supremum over polygonal partitions with rational times; the geodesic condition is the conjunction of the endpoint condition with equality between path length and endpoint distance. The regularity conditions (7)–(8) involve only countably many dyadic scales and therefore remain Borel. Hence the full witnessing set is Borel.

A projection of a Borel subset of a product of Polish spaces is analytic. Analytic sets are universally measurable, and the Jankov–von Neumann selection theorem supplies a universally measurable selector on the projected event. Projecting the selector further gives universally measurable witnessing endpoints in the compact endpoint set. Completing the underlying probability space therefore permits these endpoint variables to be adjoined to subsequence representations without any separate tightness issue for the path coordinate. Equivalently, one may formulate the argument throughout with outer probabilities. □

**Proposition 4.3** (Small-scale comparison at regular endpoints)**.** *For every compact $K \Subset U$, fixed $\alpha \in (3/4,1)$, and $\eta > 0$,*

$$\sup_{z\in K} \mathbb{P}\left[\begin{array}{c} \textit{some dyadically regular pair at } (z,r) \textit{ satisfies} \\ e(u,v) \geq (1+\eta) D_h(u,v) \end{array}\right] \longrightarrow 0 \qquad \textit{as } r \downarrow 0.$$

*Proof.* Suppose not. Choose a failing sequence $r_n \downarrow 0$, $z_n \to z \in K$. Let $h_{f,r}(z) = (h(z+r\cdot), f)$ and put

$$\widehat{h}_n = h(z_n + r_n \cdot) - h_{f,r_n}(z_n), \qquad \psi_n(w) = \frac{\phi(z_n + r_n w) - \phi(z_n)}{r_n}.$$

The fields $\widehat{h}_n$ have the same smoothly normalized whole-plane GFF law. Affine covariance and Weyl scaling identify the rescaled source and pullback metrics with the original metrics multiplied by

$$r_n^{-\xi Q} e^{-\xi h_{f,r_n}(z_n)}.$$

The maps $\psi_n$ converge locally to the affine map $w \mapsto \phi'(z)w$, and their domains exhaust $\mathbb{C}$.

Choose expanding closed disks as in Lemma 3.3 and confine both rescaled path metrics to them. Lemma 2.1 compares lengths, hence also confined metrics, with the same deterministic constant $C$. A comparison failure for the original $e$ remains a failure after confinement because confinement can only increase the $e$-distance. The specified source geodesic lies in the unit annulus, so its source distance is unchanged by the confinement.

After normalization, the upper bound (7) is a tight random multiple of $(1-\alpha)^{-1}$: the discrepancy between a circle average and the fixed smooth average has a fixed Gaussian law. Restrict first to the event that this bound is at most $M$ and then let $M \uparrow \infty$. Endpoint separation is at least $1-\alpha$. The dyadic endpoint bounds (8) supply the sequence of scales required in Lemma 4.1. All small endpoint annuli are contained in the rescaled domain for large $n$.

By Lemma 4.2, on the failure event we may choose witnessing endpoints universally measurably. Since the rescaled endpoints lie in a fixed compact annulus, their laws are automatically tight. Apply

a Skorokhod subsequence representation to the joint field/metric convergence of Lemma 3.3, together with these selected endpoint variables on the event of failure. On any represented subsequence along which failure persists and the normalized source distance is at most $M$, compactness of endpoint positions gives a further subsequence to which Lemma 4.1 applies. That lemma rules out the failure. Therefore the failure probability tends to zero. The contradiction argument was for an arbitrary failing sequence $z_n \in K$, so the convergence is uniform in $z \in K$. □

# 5 Local good annuli

We now isolate the geometric input from Ding–Gwynne [1, Section 3.2, especially Lemmas 3.13–3.18]. This prevents us from treating a nonlinear pullback as though it were itself an axiomatic weak LQG metric.

Fix the constant $C$ from Lemma 2.1 and put

$$k = \frac{1}{4C^2}.$$

For $A = A_{r/2,2r}(z)$ set

$$S = D_h(A_{3r/4,3r/2}(z), \partial A; A).$$

Given $p < 1$, the source-only estimates of [1] allow us to choose a dyadic $\delta_0 \in \Delta \cap (0, 1/100)$, then $\alpha \in (1 - \delta_0, 1)$, and $A_0 > 1$ so that, with probability at least $p$ uniformly in $z, r$, the following hold:

(G1) For every $D_h(\cdot, \cdot; A)$-geodesic between boundary points of $A$, every $\delta \in \Delta$ with $0 < \delta \le \delta_0$, and every $x \in A_{3r/4,3r/2}(z)$ whose $\delta r$-ball meets the geodesic,

$$D_h(\text{around } A_{\delta r,\sqrt{\delta} r}(x)) \le \delta^{\theta} D_h(\text{across } A_{\delta r,\sqrt{\delta} r}(x)).$$

(G2)

$$D_h(\text{around } A_{3r/2,2r}(z)) \le \min\{(1-\alpha)^{-1} m_{z,r},\ k\delta_0^{-\theta} S\}.$$

(G3) Each segment of such an internal geodesic lying in $A_{\alpha r,r}(z)$ has Euclidean diameter at most $\delta_0 r$.

(G4)

$$D_h(\text{around } A_{\alpha r,r}(z)) \le A_0 D_h(\text{across } A_{\alpha r,r}(z)).$$

The restriction to dyadic $\delta$ only weakens the corresponding source event in [1]. Its verification uses only the source metric: regularity near geodesics, the non-tracing estimate, and annular tightness. No axiom of a second metric is used.

Add the fifth condition:

(G5) Whenever a dyadically regular pair $u, v$ at $(z, r)$ satisfies

$$|u - v| \le \delta_0 r, \qquad D_h(\text{around } A_{\delta_0 r,\sqrt{\delta_0} r}(v)) \le kS, \tag{9}$$

then

$$e(u, v) \le (1 + \eta) D_h(u, v).$$

Call the resulting event $F_{z,r}$.

**Lemma 5.1** (Locality and probability of good annuli)**.** *The event $F_{z,r}$ is measurable from the field on $A_{r/2,2r}(z)$ modulo an additive constant. Given $p < 1$, choose the source geometric parameters above with sufficiently smaller failure probability. Then for every compact $K \Subset U$ and every $\eta > 0$,*

$$\inf_{z \in K} \mathbb{P}[F_{z,r}] \geq p$$

*for all sufficiently small $r$. The parameters $\delta_0, \alpha, A_0$ can be chosen independently of $\eta$.*

*Proof.* Only (G5) requires discussion. On (9), choose a surrounding source loop of length less than $2kS$. By Lemma 2.1, its $e$-length is less than

$$2CkS = \frac{S}{2C},$$

whereas an $e$-path escaping $A$ from the middle region costs at least $S/C$. The same strict inequality holds for $D_h$. Hence a path that leaves $A$ can be shortened after its first and last intersections with the loop. Therefore both distances between $u$ and $v$ agree with the distances intrinsic to $A$, and a source geodesic between them does not leave $A$. Near-minimizing paths suffice; no $e$-geodesic is needed.

All remaining regularity tests use endpoint balls and annuli compactly contained in $A$. A point-to-circle distance is determined by paths stopped at their first exit. Thus (G5) is a condition on the two internal metrics of $A$, both measurable functions of the annular field by locality; simultaneous Weyl scaling removes the additive constant. Proposition 4.3 gives the probability estimate for (G5), while the first four conditions come from the source estimates. □

Choose $p$ close enough to one to apply annular near-independence with failure exponent greater than 10. At lattice centers of mesh $\varepsilon^2/100$ in a neighborhood of $K$, test geometrically separated radii in $[\varepsilon^2, \varepsilon]$. Lemma 5.1, annular near-independence, and a union bound imply

$$\mathbb{P}\left[\forall x \in K \ \exists (z,r) : x \in B_{r/2}(z),\ r \in [\varepsilon^2, \varepsilon],\ F_{z,r}\right] \longrightarrow 1. \tag{10}$$

Choose a deterministic subsequence $\varepsilon_j \downarrow 0$ with summable failure probabilities, simultaneously for a countable compact exhaustion. Then almost surely the good covers occur along this subsequence on every compact set.

# 6 Improving the comparison constant to one

**Proposition 6.1.** *Almost surely $e \leq d$ on $U \times U$.*

*Proof.* Let

$$C_* = \inf\{c \geq 1 :\ \mathbb{P}[e(u,v) \leq c\, d(u,v) \text{ for all } u, v \in U] = 1\}.$$

Lemma 2.1 gives $C_* < \infty$, and the bound at $C_*$ holds by intersecting a countable decreasing sequence of admissible constants.

Suppose $C_* > 1$ and fix $0 < \eta < C_* - 1$. Define

$$C_{\text{new}} = (1+\eta) + \frac{A_0}{A_0 + 1}(C_* - 1 - \eta). \tag{11}$$

Then

$$C_* - C_{\text{new}} = \frac{C_* - 1 - \eta}{A_0 + 1} > 0.$$

Work on the probability-one event of good covers. Let $P : [0, T] \to U$ be a finite whole-plane $D_h$-geodesic, parametrized by length, whose image is compactly contained in $U$. At a good-cover scale, start at time $t_{j-1}$ and choose a good ball $B_{r_j}(z_j)$ with $P(t_{j-1}) \in B_{r_j/2}(z_j)$. Let $t_j$ be its next exit time and $s_j$ the last preceding time at which $P$ lies on $\partial B_{\alpha r_j}(z_j)$. Ignore the initial and terminal portions corresponding to balls meeting the endpoints of $P$; their Euclidean diameters tend to zero with the cover scale.

Restrict $P$ between its surrounding hits of the boundary of $A_{r_j/2,2r_j}(z_j)$. This is an internal geodesic to which (G1)–(G3) apply. The short outer loop from (G2) meets the extended geodesic on both sides. Minimality yields

$$t_j - s_j \le (1-\alpha)^{-1} m_{z_j,r_j}.$$

Condition (G3) gives $\operatorname{diam} P([s_j, t_j]) \le \delta_0 r_j$. Condition (G1), applied on the dyadic scales, gives exactly the dyadic endpoint estimates (8); minimality of the extended geodesic converts surrounding-loop bounds into the required endpoint-to-circle bounds. Applying (G1) at the dyadic scale $\delta_0$ and then using the second bound in (G2) yields (9). Thus (G5) gives

$$e(P(s_j), P(t_j)) \le (1+\eta)(t_j - s_j). \tag{12}$$

The defining bound at $C_*$ gives

$$e(P(t_{j-1}), P(s_j)) \le C_*(s_j - t_{j-1}). \tag{13}$$

Finally, (G4) and geodesic minimality imply

$$s_j - t_{j-1} \le A_0(t_j - s_j). \tag{14}$$

Combining (12)–(14) over consecutive intervals yields an $e$-cost at most $C_{\text{new}}$ times the retained source length.

The discarded endpoint pieces have source length tending to zero. Indeed, a nonconstant finite geodesic is injective and Euclidean continuous, hence a homeomorphism from each compact time interval onto its image. Times at which it lies in shrinking Euclidean neighborhoods of an endpoint must converge to the corresponding endpoint time. Their $e$-cost is at most $C_*$ times their source length. Sending the cover scale to zero gives

$$e(P(0), P(T)) \le C_{\text{new}} T. \tag{15}$$

The good-cover event did not depend on $P$, so (15) holds simultaneously for every such geodesic.

Now let $Q$ be an arbitrary finite $D_h$-length path in $U$. Its image is compact. Choose compact $K_0$ with $Q([0,1]) \subset \operatorname{int} K_0 \Subset U$. The local inverse-Hölder estimate gives a strictly positive lower bound for the $D_h$-distance from $Q([0,1])$ to $\partial K_0$. Partition $Q$ into finitely many pieces whose lengths are smaller than this bound. Whole-plane geodesics between consecutive endpoints exist and stay inside $K_0$, because exiting $K_0$ would already cost more than the piece they replace. Apply (15) to each replacement, sum, and infimize over $Q$. We obtain $e \le C_{\text{new}} d$ at every endpoint pair; when $d = +\infty$ the inequality is automatic. This contradicts the definition of $C_*$. Hence $C_* = 1$. □

# 7 Local field variants and the inverse map

**Lemma 7.1** (Transfer under local continuous corrections)**.** *Assume the one-sided inequality of Proposition 6.1 has been proved for restrictions of a whole-plane GFF. Then the same inequality holds for every field which is locally a GFF plus a continuous function.*

*Proof.* Fix disks $B \Subset B' \Subset U$. Decompose the given local field on $B'$ as a zero-boundary GFF plus its harmonic part and the prescribed continuous correction. Couple the zero-boundary component with the restriction of a whole-plane GFF. On the smaller disk $B$, the difference of the two fields is a random continuous function: it is the difference of their harmonic parts plus the prescribed continuous term.

The one-sided inequality for the whole-plane field holds on a probability-one event. Simultaneous continuous Weyl scaling holds, on another probability-one event, for every continuous correction at once. Hence adding the random continuous difference transfers the inequality pathwise to the given local field on $B$. Under the conformal map, the same correction is composed with the inverse map, so corresponding path weights agree exactly.

Choose a countable disk cover with rational centers and radii, each disk compactly contained in a slightly larger one, and intersect the corresponding probability-one events. Any finite-length path has compact image and can be subdivided into finitely many pieces lying in members of this cover. Adding the local inequalities over the subdivision and then infimizing gives the one-sided inequality for the intrinsic metrics of the full domain, simultaneously for all endpoint pairs. □

*Proof of Theorem 1.1.* Proposition 6.1 and Lemma 7.1 give, in the forward convention,

$$e \le d$$

for fields locally equal to a GFF plus a continuous function. Apply the same statement on $V$ to the field $h^\phi$ and the inverse conformal map $\phi^{-1}$. By the chain rule,

$$(h^\phi)^{\phi^{-1}} = \left(h \circ \phi^{-1} + Q \log |(\phi^{-1})'|\right) \circ \phi + Q \log |\phi'| = h.$$

The inverse-map inequality is therefore $d \le e$. Hence $d = e$ as extended-valued metrics, simultaneously for every endpoint pair. Rewriting this equality in the inverse convention gives (1). □

## Self-audit note

This revision was prepared after an adversarial review of the previous draft. In particular: (i) the open-domain lower-semicontinuity shortcut was replaced by compact confinement; (ii) the endpoint regularity condition was reduced to countably many dyadic scales; (iii) the measurable-selection step in the regular-pair argument was stated as an explicit analytic-set lemma; (iv) the uniformity in the rough comparison was tied to the normalized compact family of univalent maps; and (v) the local-field extension was separated into its own lemma. This note records internal verification only; it is not a claim of external referee certification.

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